

Avian Population Estimation using Passive Acoustic Monitoring

Darukaa.earth technical challenge | BioDCASE 2026 Bird Counting | Anisha Tiwary | August 2026

Short version. My final model gets **MAE 45.32** on held-out aviaries, compared to **59.07** for a baseline that ignores the audio completely. That looks like a 23% win. But I ran a permutation test to check, and it came back **p = 0.385**, so I can't tell that improvement apart from luck. I'm not going to claim the audio features work. What I can show is why the dataset makes this so hard, and what would actually be needed to answer the question properly.

1 The problem, and the thing that makes it hard

The task is counting, not detecting. I get about 20,000 three-second clips from a zoo aviary and have to output one number: how many birds of the target species live there. The label comes from zoo records, so it describes the whole collection, not any single clip.

Here is the problem. The development set has 140,899 clips but only **eight labels**. And two of those aren't independent: **dev_aviary_5** and **dev_aviary_6** are the same 52 birds in the same place, just recorded on different days. So really there are **five independent sites**. Split by species it gets worse: four flamingo points, two quelea, two ibis. Fitting a separate model per species would mean drawing a two-parameter line through two points, which tells you nothing.

So the thing limiting me isn't how well I can detect birds. It's how complex a model I'm allowed to fit. A BirdNET or ARIA embedding gives me around 1,024 numbers per clip. Anything flexible enough to use that will fit eight points perfectly and then fail completely on a new aviary. Section 4 is where I actually test that instead of just asserting it.

2 How the pipeline works

Stage 1: measure every clip, without training anything. For each clip and each species' frequency band, I compute seven numbers from a spectrogram (64 ms window, 16 ms hop, after downsampling to 16 kHz): the peak level in the band, the 10th-percentile *floor* level, how much of the clip is near peak loudness, the onset rate, spectral flatness, how many spectral peaks appear at once, and an SNR measured against a frequency range outside the band. None of this is learned. If I trained a per-clip classifier against eight aviary-level labels, it would just learn to recognise each aviary's room acoustics and background hum, and I'd have no way to tell that apart from it learning about birds.

Stage 2: build features that don't max out. The obvious feature is "what fraction of clips have a call in them". It's useless here. Once you have roughly thirty flamingos calling, it sits at 1.0 and stays there, so 52 birds and 161 birds look the same. I tried two ideas to get past that. The **chorus floor**: in a big flock, other birds fill in the gaps between one bird's calls, so the quiet part of each clip gets louder even after the loud part has maxed out. And **polyphony**: more birds calling at once should make the spectrum flatter and busier. Every loudness-based feature is measured against that aviary's own quiet baseline, so differences in recording gain between sites cancel out. Twenty features total per aviary and species.

Stage 3: keep the model tiny. $\log N = \alpha_{\text{species}} + \beta \cdot z(\text{feature})$. Three species intercepts plus one shared slope, so four parameters over eight points. The log link is because populations run from 4 to 161 and being 10 out on a pair of ibis is not the same as being 10 out on a flock of 161. Sharing one slope across species means all eight points help estimate it, instead of three separate hopeless fits. And I constrain the slope to be non-negative, because "more calling means fewer birds" is never a real relationship.

3 How I evaluated it

- **Leave one aviary out**, five folds, with aviaries 5 and 6 held out together so the model can't learn one site's acoustics and apply it to the same site under a different name.
- **Feature selection inside each fold.** With 20 features and 8 points, if I pick the best feature using all the data and then report its cross-validation score, that score is meaningless. I can always find something that fits. So selection runs inside each fold, and I report both the honest number and the cheating one so you can see the gap.
- **The baseline to beat:** just predict the average count for that species from the training folds, using no audio at all. It's a strong baseline because the task already tells you the species.
- **Data:** I used 400 clips per aviary (1 to 3% of each collection), spread evenly across every hour of every day rather than taken as one block, so daily activity patterns survive. This was a time trade-off. It's defensible because every feature is a rate or a percentile, so none of them depend on how many clips I sampled.

4 Result 1: fewer parameters worked much better

I ran the pipeline twice. The only difference is how many features the model is allowed to use. Same data, same features available, same evaluation.

Model	MAE	RMSE	MAPE %
Baseline, no audio at all	59.07	70.24	85.9
Run A, 2 features (honest)	93.34	120.84	205.0
Run B, 1 feature (honest)	45.32	55.91	115.9
Run B, 1 feature (cheating)	42.12	51.48	109.8

Run A has more freedom and does worse than using no audio at all.

Cutting from two features to one more than halved the error. That's the clearest evidence I have that model size, not feature quality, was what was holding things back. You can see the same thing in which features got picked: Run A chose a different pair in every single fold, while Run B picked `bout_burstiness` in four folds out of five. Run A's cheating score is also *better* than its honest one (48.2 against 93.3), which is what it looks like when feature selection is just fitting noise.

Same birds, different day

Aviaries 5 and 6 are the same 52 birds recorded on separate days. So the gap between the two predictions tells me directly how much of my answer comes from the recording conditions rather than the birds. It's the cleanest check available in this dataset and it costs nothing to run.

Configuration	Aviary 5	Aviary 6	Ratio
Run A (2 features)	184.0	313.4	1.70x
Run B (1 feature)	132.6	133.2	1.005x

Run A calls the same flock 184 birds one day and 313 the next. Run B agrees with itself to within half a percent. Worth being clear though: Run B is *consistently wrong*. It says about 133 when the answer is 52. Being reproducible and being accurate are two different things, and right now I only have the first one.

The sign constraint did more work than I expected

Seventeen of my twenty features scored exactly 59.07, which is the baseline value to two decimal places. That's not a coincidence. Those features all came out with a negative slope, the constraint clipped them to zero, and the model fell back to just predicting the species average. Without that one line, each of them would have added a fake "more noise, fewer birds" relationship that fits my eight training points and flips on anything new.

5 Result 2: I checked whether the improvement was real

With 8 points and 20 features, does beating the baseline by 23% actually mean anything? To find out I shuffled the eight count labels, re-ran the whole pipeline including feature selection, and repeated that 500 times. If random labels often do as well as my real ones, then my result is noise.

What I shuffled	Null MAE median	Null 5th pct	p
Counts shuffled freely	81.31	37.89	0.080
Counts shuffled within each species	50.90	31.24	0.385

The second row is the one that matters. Shuffling freely mostly just destroys the species information, which I already know is useful. Shuffling within species keeps that intact and only breaks the link between the features and the counts, so it isolates what the audio is actually adding.

It failed. My 45.32 is better than the median of 50.90, but random labels do at least as well 39% of the time. I can't call that a real effect.

Is this just because the dataset is tiny? Largely, yes, and it's worth spelling out. Shuffling within species means the number of possible label arrangements is $2! \times 2! \times 4! = 96$. That's it. So the smallest p-value any experiment of this shape could ever produce on this data is about 0.010, and realistically the resolution is much coarser than that. Even a perfect feature would have trouble reaching significance here. That doesn't save my result, a failed test is a failed test. But it does tell me what's actually missing: **more labelled sites, not more audio**. Downloading another million clips from these six aviaries would not move this number at all.

6 Result 3: I made my own ground truth

Eight labels isn't enough to test whether a feature tracks abundance. So instead of accepting that, I built labels. I take real clips from **dev_avuary_4**, add K of them together, and measure how each feature responds as K goes from 1 to 64. That gives me thousands of exactly-labelled points instead of eight, using real audio with real background noise and room acoustics.

One thing I had to be careful about. Adding K signals together raises the overall level by about the square root of K, which would make any loudness feature go up for trivial reasons. So I compute every curve twice, once raw and once with the mix rescaled to constant loudness, and I only draw conclusions from the rescaled version. There's a second smaller issue: the background noise adds up along with the calls, so my saturation points are probably a bit optimistic.

K (sources mixed)	1	4	8	16	32	64
peak_db	-17.4	-19.5	-19.8	-20.1	-20.7	-20.9
floor_db	-34.1	-31.0	-29.0	-28.8	-28.2	-27.5
occupancy	0.14	0.32	0.46	0.48	0.66	0.71
n_peaks	16.2	16.4	16.4	16.5	16.4	16.8

Loudness-normalised medians, flamingo band 350 to 2200 Hz, 40 trials per K.

- **The chorus floor idea was right.** With loudness held constant, peak_db stays flat the whole way while floor_db climbs steadily by **6.6 dB** and still hasn't levelled off at K = 64. That's the biggest range of any feature I measured. The energy is moving out of the peaks and into the gaps, exactly as I'd predicted. occupancy shows the same thing from the other direction, going from 0.14 up to 0.71.
- **Counting events is hopeless.** onset_rate stops responding at **K = 6**. Every flock in this dataset is way past six birds. This is the strongest argument I found against building a counter on top of a detector.
- **One of my assumptions was just wrong.** n_peaks barely moves (16.2 to 16.8). Counting spectral peaks does not tell you how many birds are calling, even though I included it specifically for that.

So why did the chorus floor fail on the real data? This experiment says chorus floor and occupancy are good features that behave exactly as designed. On the actual eight-label task they were some of the worst. Both results are correct, and I think the explanation is this: my mixing experiment holds the site, the microphone, the background and the other species constant and only changes density. The real task changes all of those at once across five different sites. A feature can respond beautifully to density and still fail as an estimator, because the differences between sites drown out the signal inside each site. **The chorus floor idea isn't wrong. It's buried.**

7 What went wrong, and where

Measuring a frequency band is not the same as identifying a species. This is my biggest problem. In **dev_avuary_5**, which has *no* Red-billed quelea in it at all, the quelea band (2 to 8 kHz) has an activity rate of 0.795. That's higher than the flamingo band's 0.568 in the same aviary, where 52 flamingos actually live. My measurement reports how much energy is in a frequency range. It has no way of knowing what made that energy. Each aviary has between 2 and 12 other species in it, so every feature is contaminated by an unknown amount. This is very likely why the chorus floor features failed.

Flamingos: maxed out, as expected. Errors of +6, -30, +81, +81 against true counts of 161, 107, 52, 52. Everything gets squashed toward the average. The features I built to fix this didn't survive selection, so nothing in the final model handles it.

Quelea: backwards. 153 birds predicted as 73, and 61 predicted as 127. The aviary with more birds got the lower number. With only two quelea points held out one at a time, each prediction is basically extrapolated from the other species. That's a consequence of how little data there is, not a problem with the feature.

Ibis: wrong scale entirely. Counts of 4 and 6 predicted as 2.0 and 18.3. That second one is 358% off, and it's why Run B improves MAE but makes MAPE worse (115.9 against 85.9). A model fitted mostly on flocks of 52 to 161 can't tell the difference between four birds and six.

One feature that did nothing. diel_active_hours came out as 24.0 for every aviary and every band, because background noise alone clears the activity threshold at all hours. I left it in the code so it's visible, but it should be redefined against a relative threshold instead of a fixed one.

8 What I'm not claiming

The main result isn't statistically significant ($p = 0.385$), and given the dataset no version of this experiment could have got below $p = 0.010$. The frequency bands come from the literature, they're assumptions rather than measurements, and section 7 shows they don't cleanly separate species. The feature my final model uses, `bout_burstiness`, measures timing between active clips, which makes it more sensitive to my 400-clip subsample than the other features are. Subsampling imposes its own timing structure, and I haven't rechecked this at fuller sampling. My mixing experiment adds backgrounds together along with the calls and assumes birds call independently, when real flocks partly synchronise. Pied avocet has no development label so its predictions use a fallback intercept and are marked `extrapolated=1`.

And the assumption underneath all of it: **calling is behaviour, not a headcount**. A quiet bird is invisible to any acoustic method. Everything here estimates how many birds are *calling*, and turning that into a population number is its own separate assumption that I haven't tested.

9 What I'd do next

- **Get more labelled sites.** The limit here is eight aggregate points, not hours of audio. Something like 15 to 20 independently labelled aviaries would give this design enough resolution to detect an effect the size of the one I saw. No amount of modelling cleverness substitutes for this.
- **Use a detector to gate the band measurements.** Run BirdNET or ARIA over a subsample, not to generate features, but to confirm the target species is actually present and check my frequency bands are right. Then only measure frames where it was detected. This costs nothing in model complexity because the detector is validating, not predicting. Section 6 already showed the underlying mechanism is real, so this is a targeted fix rather than a guess.
- **Cluster individual calls for the small populations.** For ibis and avocet, take embeddings of isolated calls, cluster them, and count the clusters. It's the only approach that could tell four birds from six, and small populations are exactly where it's computationally feasible.
- **Turn the mixing experiment into the actual model.** Extend section 6: build chorus at known N from single-bird recordings, measure how the chorus floor grows, and invert that curve. This swaps eight real labels for an unlimited synthetic calibration set and makes the problem physical rather than statistical.
- **Try leaving a whole species out.** Fit on flamingo and quelea, then predict ibis. If the shared slope is real biology it should carry across two orders of magnitude. Much harder than leaving one aviary out, and failing it would tell me something useful.

10 Running it

`bash run_all.sh` reproduces everything from a fresh clone, with all random seeds fixed. `python scripts/smoke_test.py` runs every stage on generated audio so you can check the setup works without downloading anything, which matters because the full dataset is around 287 GB. After the download it all runs in under ten minutes on four cores. There's no deep learning in the default path, the measurement code is plain scipy.

What	Where
Clip measurement, features, model, inference	<code>src/</code>
Permutation test	<code>scripts/permutation_test.py</code>
Mixing / saturation experiment	<code>scripts/saturation_experiment.py</code>
End-to-end test on synthetic audio	<code>scripts/smoke_test.py</code>
Longer version of this report	<code>report/TECHNICAL_REPORT.md</code>
Metrics, model, predictions, figures	<code>results/, figures/</code>

Where I ended up. I built a working pipeline, and then spent most of the time left trying to work out whether its main number meant anything. It doesn't, at least not at any level I'd defend, and the permutation arithmetic says no experiment on this development set could have shown otherwise. The mixing experiment does show the chorus floor idea is sound, which turns the negative result into a contamination problem with a specific fix rather than a dead end. If there's one thing I'd want read from this, it's what these eight labels can't support, and what it would take to get further.

Dataset: BioDCASE 2026 Bird Counting (CC BY 4.0), Argente, Härmä and Arslan-Dogan, Maastricht University.